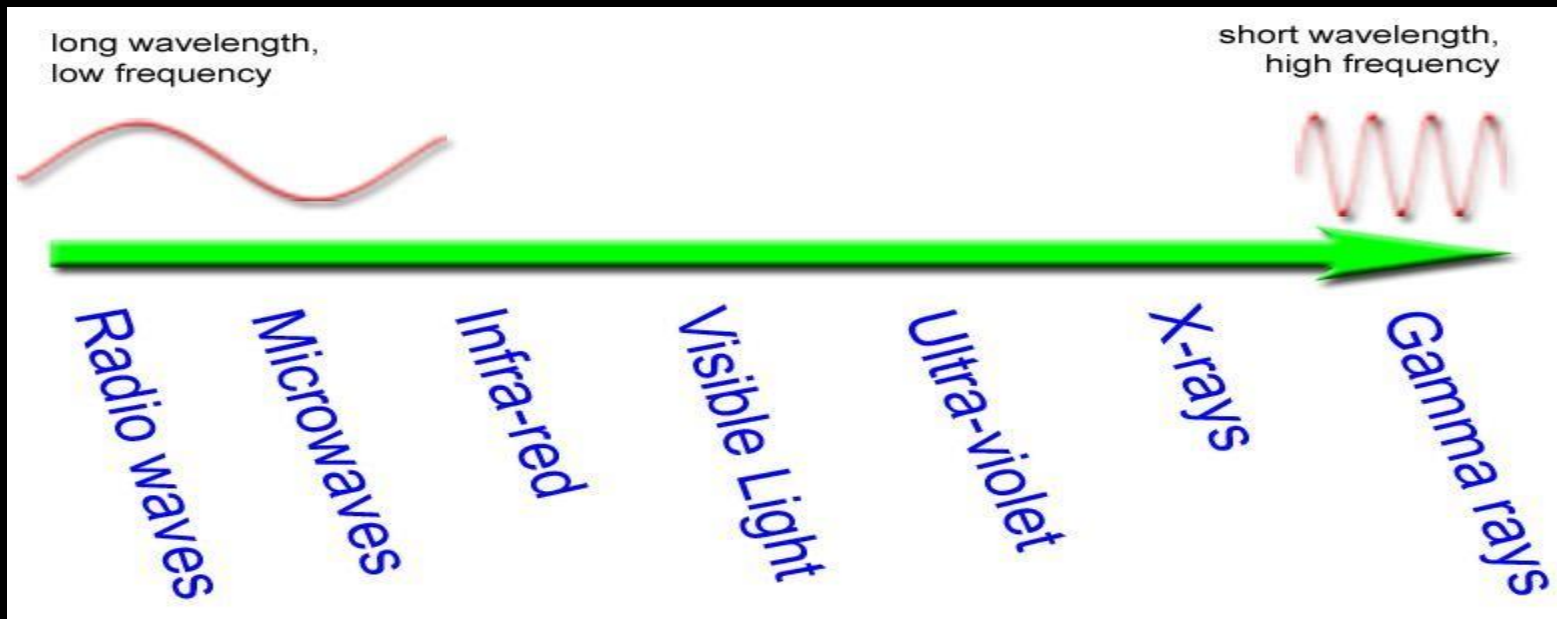


ADVANCE DIGITAL IMAGE PROCESSING

LECTURE # 2

APPLICATIONS OF IMAGE PROCESSING

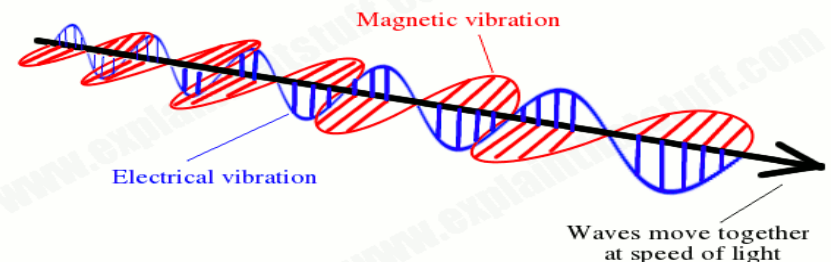
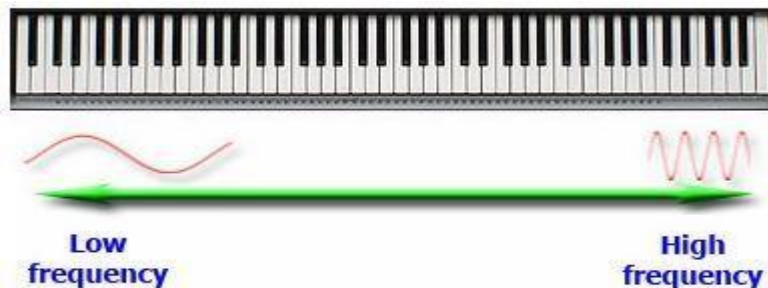
EM Spectrum



EM Spectrum [1]

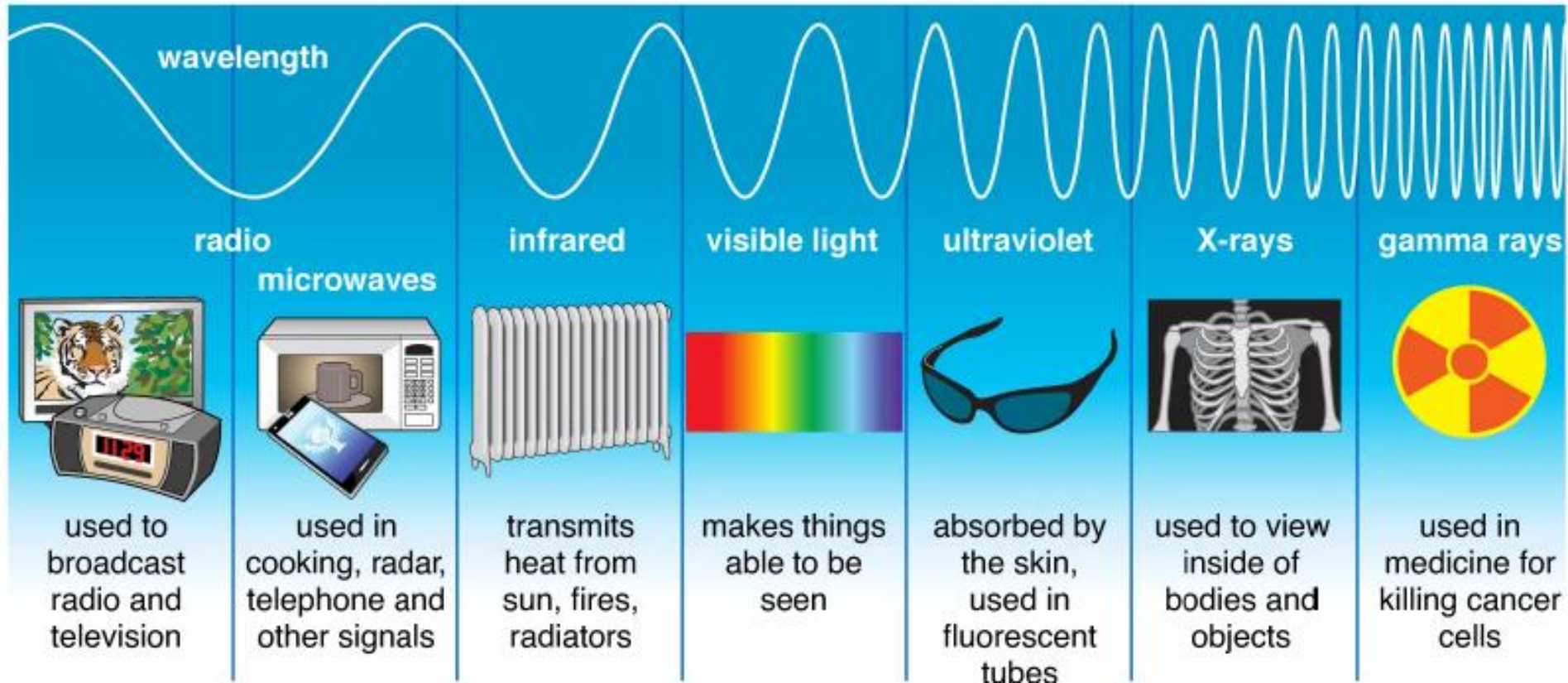
4

- ❑ Scientists have found that many types of wave can be arranged together like the notes on a piano keyboard, to form a scale.
- ❑ We can specify waves through frequency and wavelength
- ❑ The 'low notes' have a low frequency and a long wavelength.
- ❑ The 'high notes' have a high frequency and a short wavelength.
- ❑ With a sound wave, it's the air particles that are vibrating.
- ❑ So what's vibrating when an electromagnetic wave passes by?
- ❑ Electromagnetic waves are vibrations of magnetic and electric fields. So they don't need air in order to travel. They don't need anything to be there at all.



EM Spectrum [1]

Types of Electromagnetic Radiation



Gamma Rays [2]

6

- ❑ Gamma rays are given off by **stars**, and by some **radioactive substances**.
- ❑ They are extremely high frequency waves, and carry a large amount of energy.
- ❑ They pass through most materials, and are quite difficult to stop - you need lead or concrete in order to block them out.
- ❑ Because Gamma rays can kill living cells, they are used to **kill cancer cells** without having conducting difficult surgery.
- ❑ This is called "**Radiotherapy**", and works because cancer cells can't repair themselves like healthy cells can when damaged by gamma rays. Getting the dose right is very important!

X-Rays [3]

8



- ❑ X-rays are very high frequency waves emitted by stars and some types of nebula, and carry a lot of energy. They will pass through most substances, and this makes them useful in medicine and industry to see inside things.
- ❑ An **X-ray machine** works by firing a beam of electrons at a "target". If we fire the electrons with enough energy, X-rays will be produced.
- ❑ X-rays are used by doctors to **see inside people**. They pass easily through soft tissues, but not so easily through bones.
- ❑ Lower energy X-Rays don't pass through tissues as easily, and can be used to scan soft areas such as the brain
- ❑ X-Rays are also used in **airport security** checks, to see inside your luggage.



Ultraviolet ^[4]

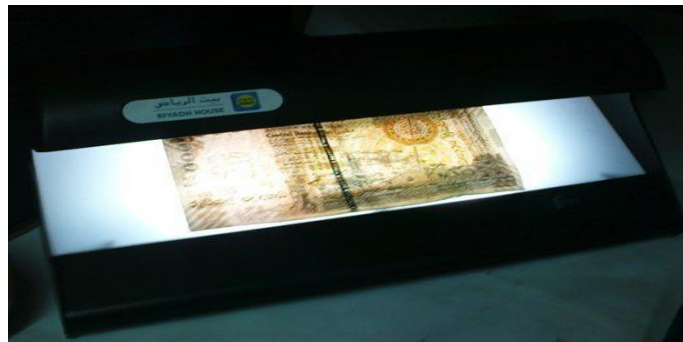
9

- ❑ **Ultraviolet (UV)** light is electromagnetic radiation with a wavelength shorter than that of visible light, but longer than X-rays, in the range 10 nm to 400 nm.
- ❑ It is named because the spectrum consists of electromagnetic waves with frequencies higher than those that humans identify as the color violet.
- ❑ These frequencies are invisible to humans, but visible to a number of insects and birds.

Uses of Ultraviolet

10

- ❑ Uses for UV light include getting a **sun tan [11]**, **detecting forged bank notes [12]**, and hardening some types of dental filling.
- ❑ Hospitals use UV lamps to sterilize surgical equipment and the air in operating theatres. Food and drug companies also use UV lamps to **sterilize** their products.[10]
- ❑ Suitable doses of Ultraviolet rays **cause the body to produce vitamin D**, and this is used by doctors to treat vitamin D deficiency and some skin disorders.



Visible Light [5]

11

- ❑ Our eyes can detect only a **tiny** part of the electromagnetic spectrum, called **visible light**.
- ❑ This means that there's a great deal happening around us that we're simply not aware of, unless we have instruments to detect it.
- ❑ Light waves are given off by **anything that's hot enough to glow**.
- ❑ This is how light bulbs work - an electric current heats the lamp filament to around 3,000 degrees, and it glows white-hot.
- ❑ The surface of the Sun is around 5,600 degrees, and it gives off a great deal of light.



Infrared [6]

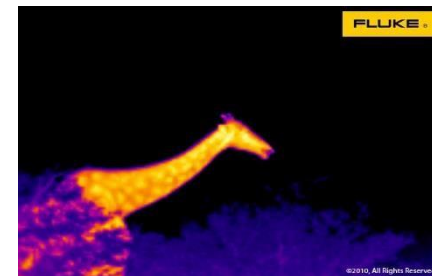
12

- ❑ Infra red waves are just below visible red light in the electromagnetic spectrum ("Infra" means "below").
- ❑ You probably think of Infra-red waves as heat, because they're given off by **hot objects**, and you can feel them as warmth on your skin.
- ❑ Infra Red waves are also given off by **stars, lamps, flames** and anything else that's warm - including you.
- ❑ The detector on the security light picks up the Infra red radiation from your body.

Uses of Infrared

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- They are used for many tasks, for example, **remote controls for TVs** and video recorders, short-range communications between mobile phones.
- Because every object gives off IR waves, we can use them to "see in the dark"
- **Night sights** for weapons sometimes use a sensitive IR detector. Remember the movie, "Predator"?
- One of the common uses for IR is in the area of security. "Passive Infra-Red" **detectors** are used in burglar **alarm systems** that many people have fitted outside their houses for security. These detect the Infra-Red emitted by people and animals.
- You've probably seen TV programs in which **police helicopters** track criminals at night, using "**thermal imaging**" cameras which can see in the dark. These cameras use Infra-Red waves instead of "ordinary" light, which is why people look bright in these pictures. Similar cameras are also used by fire crews and other **rescue workers**, to find people trapped in rubble.



TeraHertz

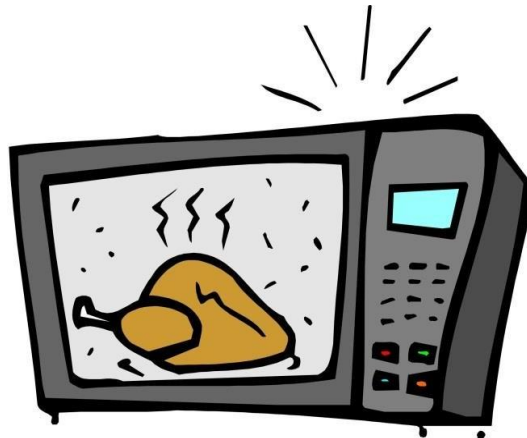
14

- ❑ Terahertz radiation is a region of the spectrum between far infrared and microwaves.
- ❑ Until recently, the range was rarely studied but applications such as imaging and communications are now appearing.
- ❑ Terahertz radiation can penetrate fabrics and plastics, so it can be used in surveillance, such as security screening, to uncover concealed weapons on a person, remotely.
- ❑ Terahertz radiation is strongly absorbed by atmospheric gases, making this frequency range useless for long distance communication.

Microwaves [7]

15

- ❑ Microwaves are basically extremely high frequency radio waves, and are made by various types of **transmitter**.
- ❑ In a mobile phone, they're made by a transmitter chip and an antenna, in a microwave oven they're made by a "magnetron".
- ❑ Stars also give off microwaves.
- ❑ Microwaves cause water and fat molecules to vibrate, which makes the substances hot. So we can use microwaves to **cook** many types of food.



Uses of Microwaves

16

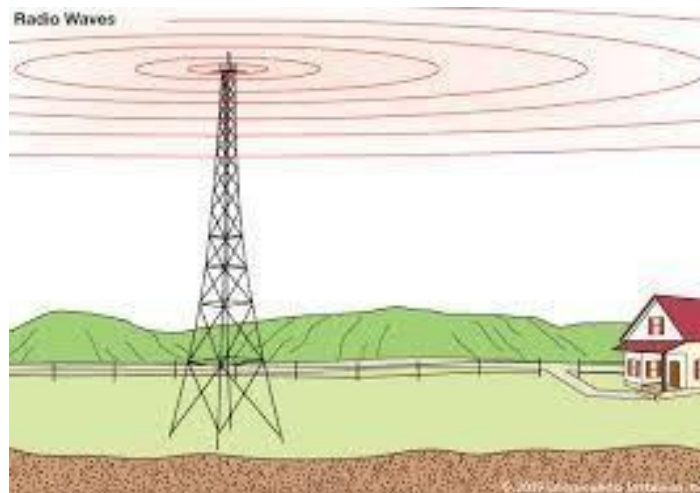
- ❑ **Mobile phones** use microwaves
- ❑ Microwaves are also used by fixed traffic **speed cameras**, and for **radar**, which is used by aircraft, ships and weather forecasters.
- ❑ The most common type of radar works by sending out bursts of microwaves, detecting the "echoes" coming back from the objects they hit, and using the time it takes for the echoes to come back to work out how far away the object is.



Radio Waves [8]

17

- ❑ Radio waves are made by various types of **transmitter** depending on the wavelength.
- ❑ They are also given off by stars, sparks and lightning, which is why you hear interference on your radio in a thunderstorm.
- ❑ Radio waves are the lowest frequencies in the electromagnetic spectrum, and are used mainly for **communications**.



Why do we Process Images?

18

- ❑ Improvement of Pictorial information for human perception
- ❑ Image Processing for Autonomous Machine
- ❑ Facilitate picture storage and transmission
- ❑ Extract information from images
- ❑ Prepare for display or Printing

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Human Perception



Human perception

Understanding why we see what we do

Human Perception



20

- ❑ Employ methods capable of enhancing pictorial information for human interpretation and analysis

- ❑ Typical Applications are::
 - ✓ Noise Filtering
 - ✓ Content Enhancement
 - Contrast Enhancement
 - De-blurring
 - ✓ Medical Imaging
 - ✓ Remote Sensing

Noise Filtering

21

- Noise can be removed from the image by using the following filters
 - ✓ Linear Filters (Average Filters, Weighted average Filters)
 - ✓ Non- Linear Filters (Order Statistics Filters [Median, Minimum, Maximum])

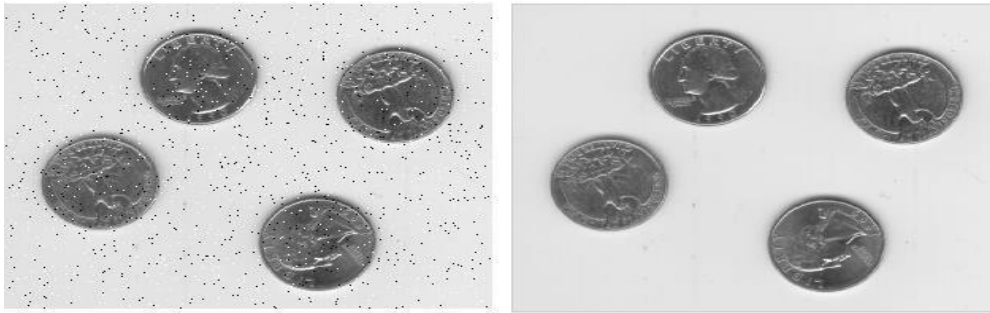
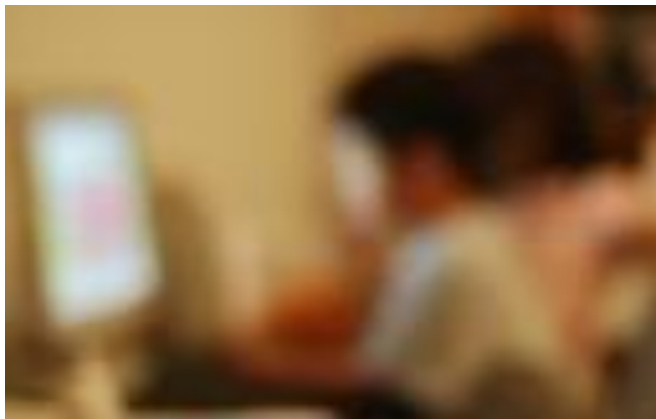


Image Deblurring

22

- There are two primary kinds of blur in photos : focus blur and motion blur.
- Here are examples of focus and motion blur, respectively:
 - ✓ Focus blur happens when the subject of your photo is simply out of focus. The solution is to make sure your autofocus is on and try again.
 - ✓ Motion blur, on the other hand, doesn't happen because your subject is out of focus. It happens because your subject is moving relative to the camera frame while the exposure is being made



Focus Blurred Image

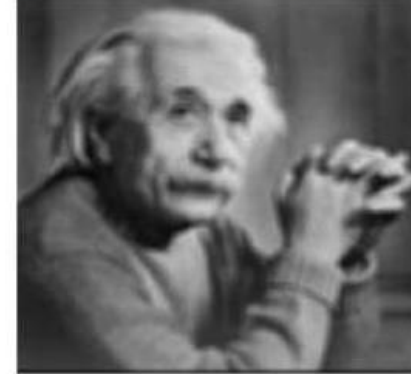
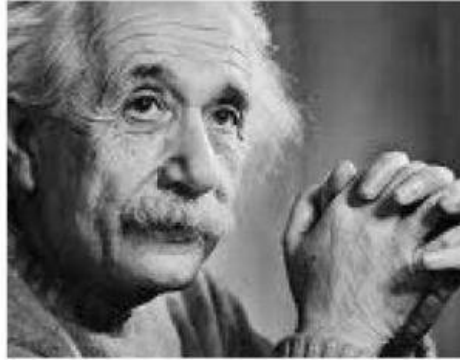
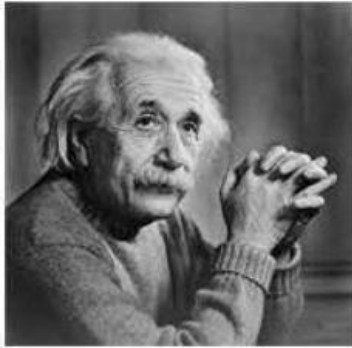


Motion Blurred Image

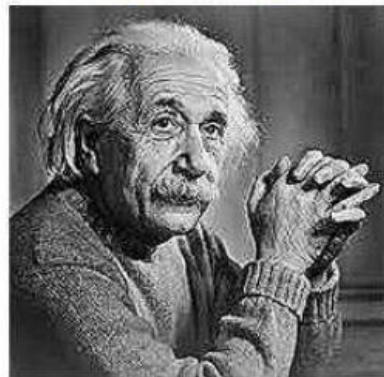
Image Sharpening

23

The original image The zoomed image Blurr image



Sharp image



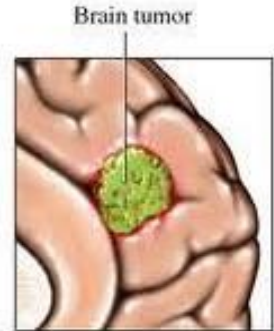
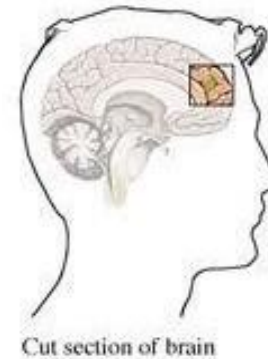
Edges



Medical Imaging

24

- Image processing has been widely used in the field of medical
- Applications of Image processing in medical are:
 - ✓ Tumor Detection
 - ✓ Cancer Detection
 - ✓ Ultrasound and many more.....



Remote Sensing

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- ❑ Remote sensing can be defined as any process whereby information is gathered about an object, area or phenomenon without being in contact with it.
- ❑ Our eyes are an excellent example of a remote sensing device. We are able to gather information about our surroundings by gauging the amount of the visible light reflected from some external source as it reflects off objects in our field of view.
- ❑ Contrast this with a thermometer, which must be in contact with the phenomenon it measures, and thus is not a remote sensing device.

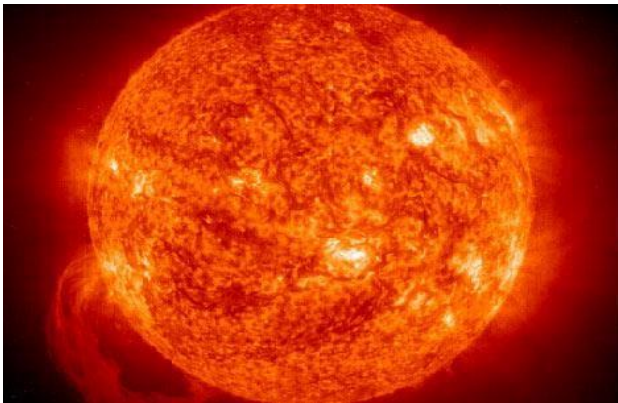


Remote Sensing

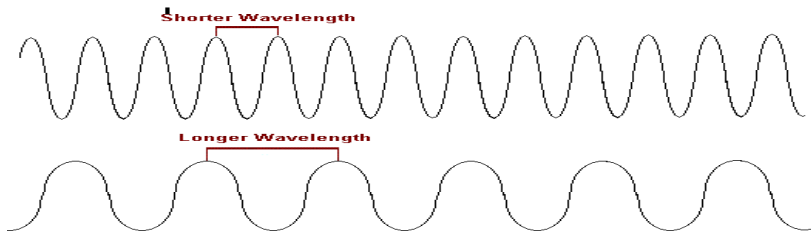
26

□ Fundamental Considerations

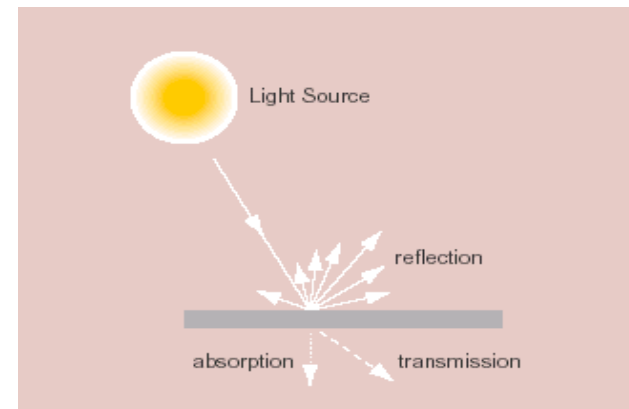
Energy Source



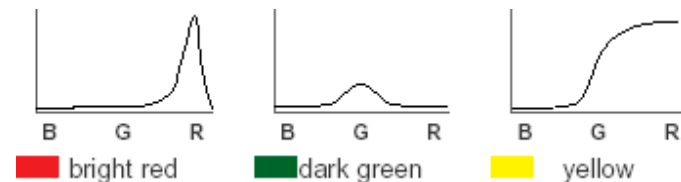
Wavelength



Interaction Mechanisms



Spectral Response Patterns



Remote Sensing- Energy Source

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- ❑ Sensors can be divided into two broad groups— *passive* and *active*.
- ❑ Passive sensors measure existing sources of energy. The majority of remote sensing is done with passive sensors, for which the sun is the major energy source.
- ❑ By contrast, active sensors provide their own source of energy. The most familiar form of this is flash photography.

Remote Sensing- Wavelength

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- ❑ As indicated, most remote sensing devices make use of electromagnetic energy. However, the electromagnetic spectrum is very broad and not all wavelengths are equally effective for remote sensing purposes. Furthermore, not all have significant interactions with earth surface materials of interest to us.
- ❑ The atmosphere itself causes significant absorption and/or scattering of the very shortest wavelengths. In addition, the glass lenses of many sensors also cause significant absorption of shorter wavelengths such as the ultraviolet (UV).
- ❑ As a result, the first significant *window* opens up in the visible wavelength.

Remote Sensing- Interaction Mechanism

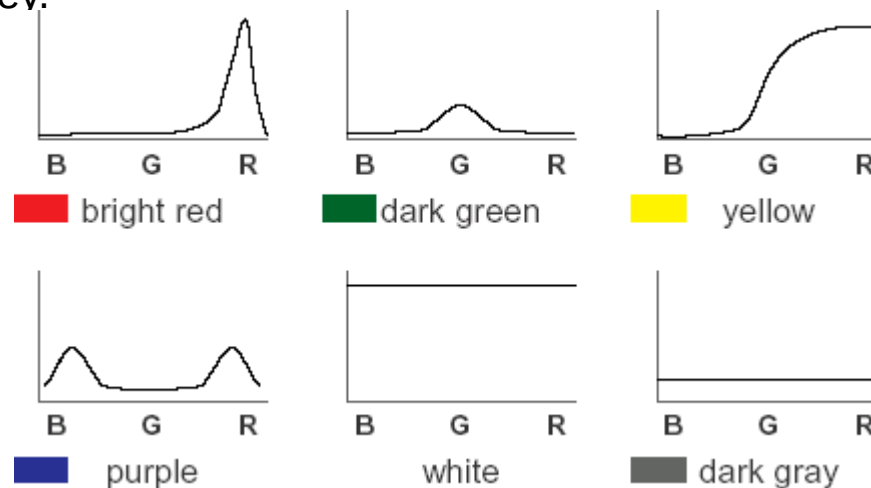
30

- ❑ When electromagnetic energy strikes a material, three types of interaction can follow:
 - ✓ reflection,
 - ✓ absorption and/or
 - ✓ Transmission
- ❑ Our main concern is with the reflected portion since the reflected part is returned to the sensor system.
- ❑ Exactly how much is reflected will vary and will depend upon the nature of the material and where in the electromagnetic spectrum our measurement is being taken.
- ❑ As a result, if we look at the nature of this reflected component over a range of wavelengths, we can characterize the result as a *spectral response pattern*

Remote Sensing- Spectral Response Pattern

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- A spectral response pattern is sometimes called a *signature*. It is a description (often in the form of a graph) of the degree to which energy is reflected in different regions of the spectrum.
- Most humans are very familiar with spectral response patterns since they are equivalent to the human concept of color.
- For example, Figure below shows idealized spectral response patterns for several familiar colors in the visible portion of the electromagnetic spectrum, as well as for white and dark grey.



Machine Vision Applications

JADAK Vision
Software



Embedded / OEM
Smart Cameras



Fixed Mount
Smart Cameras



JADAK Vision
Services & Solutions



Machine Vision

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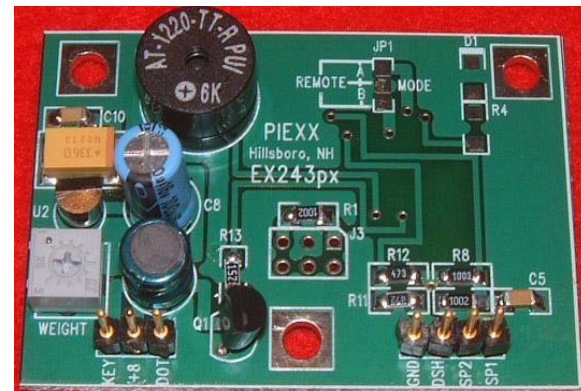
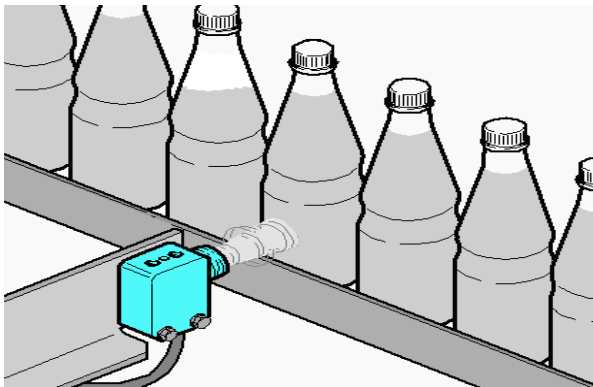
- ❑ Here the interest is on the procedures for extraction of image information suitable for computer processing
- ❑ Typical applications are:
 - ✓ Industrial Machine Vision for Product Assembly and Inspection
 - ✓ Automated Target Detection and Tracking
 - ✓ Finger Print Recognition
 - ✓ Iris Recognition
 - ✓ Face Detection, Tracking and Recognition e.t.c



Automated Product Inspection

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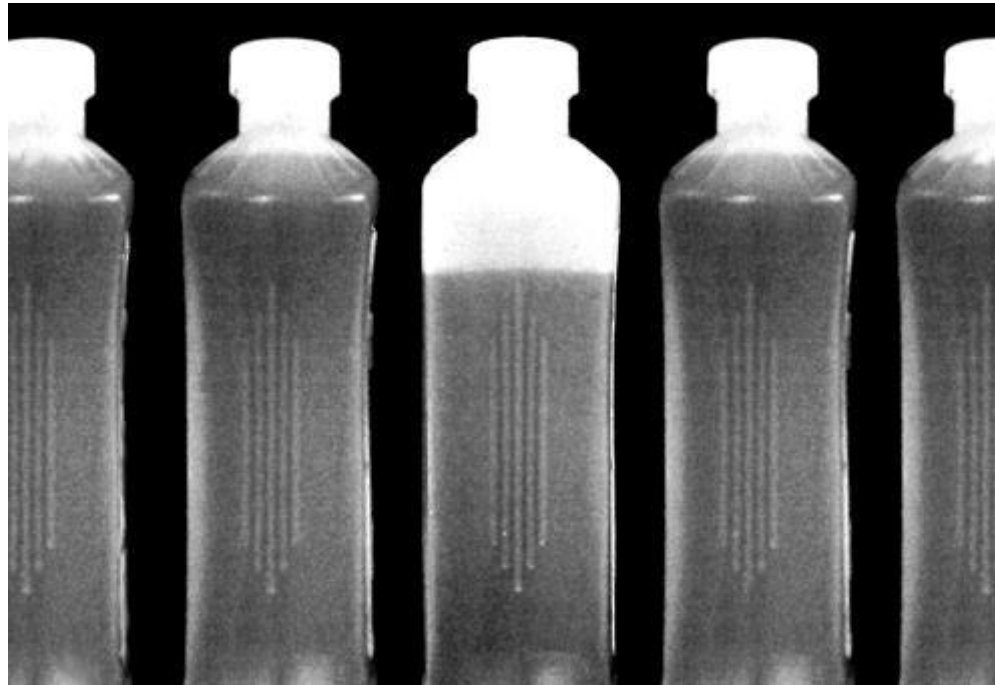
- ❑ Image Processing techniques have great utilization in the industry for various product inspections
- ❑ Some of them can be:
 - ✓ Automated Inspection of Bottling Plant Automation
 - ✓ Automated Inspection of IC Manufacturing
 - ✓ Automated Inspection of Computer Components e.t.c



Automated Inspection of Bottling Plant Automation

38

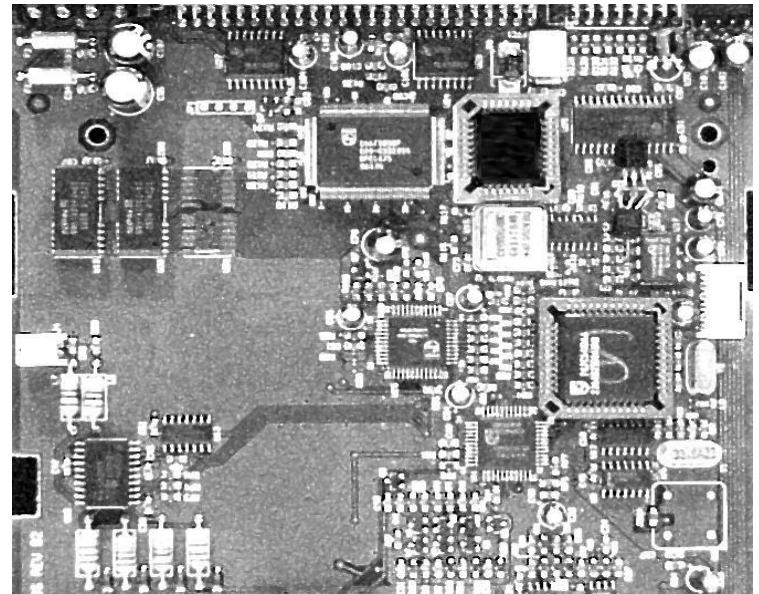
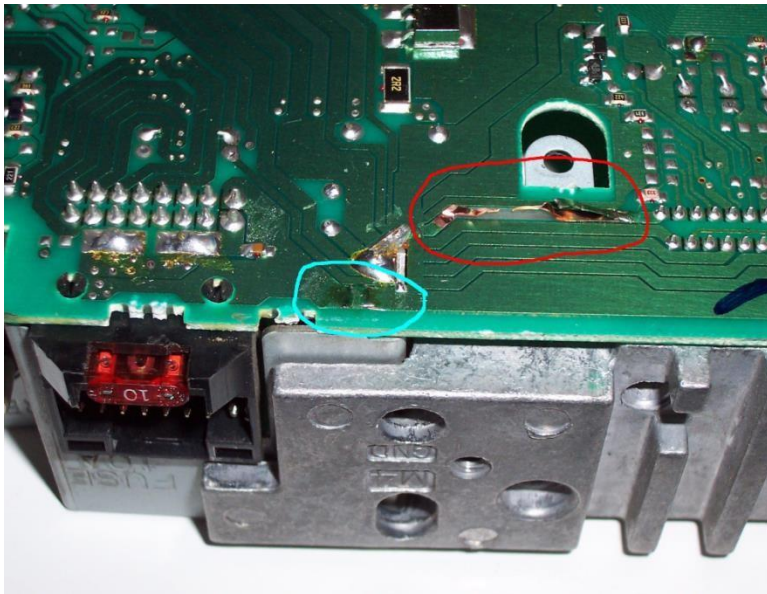
- ❑ Image processing techniques can be used to inspect the bottles of soft drinks to check whether any bottle is empty or partially filled so to avoid any such product delivered to the customer which can effect the goodwill of the company



Automated Inspection of Integrated Circuits & Circuit board/ Motherboard

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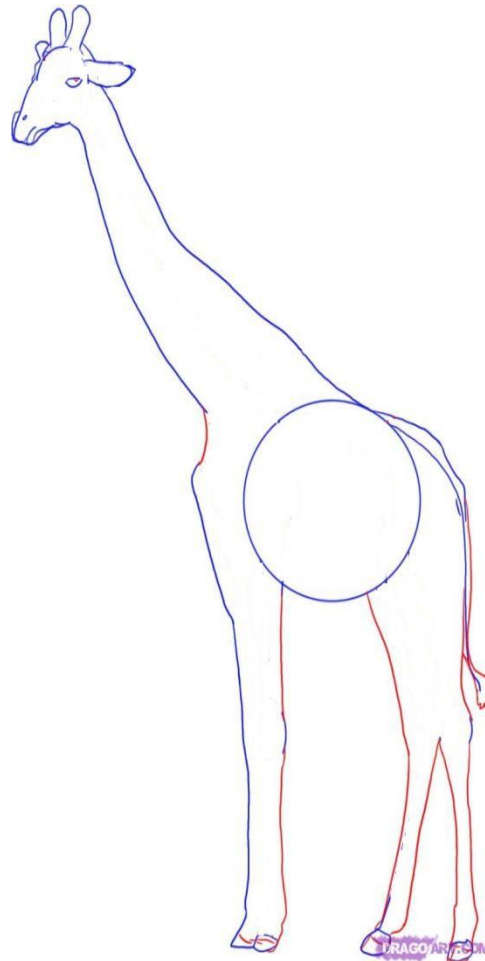
- Image processing techniques can be used to inspect the Integrated Circuits during the manufacturing phase to detect any missing components or any parts that is broken
- e.g In 1st figure some part is broken while in second figure a component is missing so these problems can be identified through image subtraction method



Boundary Information

40

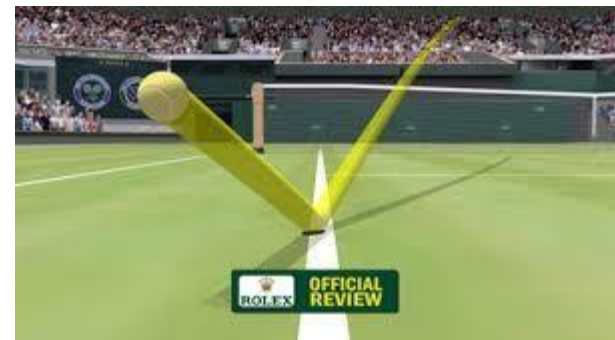
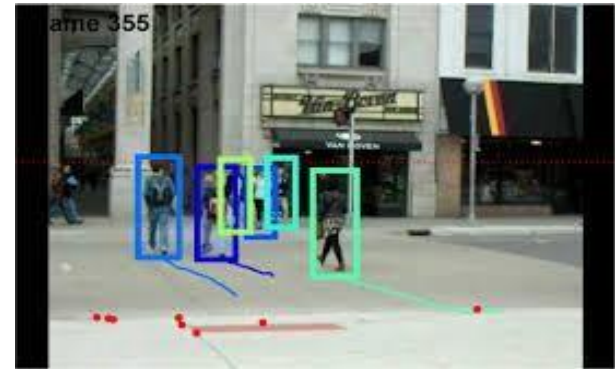
- The boundary of an object is very useful to recognize the object



Video Sequence Processing

41

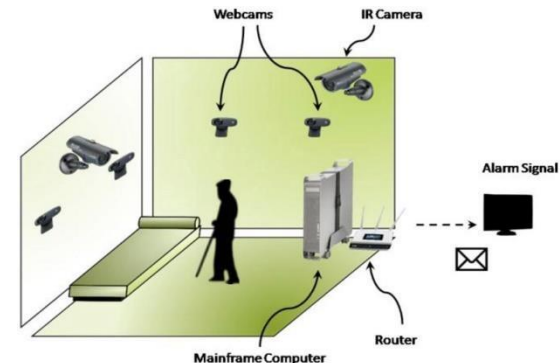
- ❑ The major emphasis of image sequence processing is detection of moving parts
- ❑ Typical applications are:
 - ✓ Detection and tracking of moving targets for security surveillance purpose
 - ✓ To find out the trajectory of a moving target
 - ✓ Monitoring of human organ for medical application



Movement Detection and Tracking

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- ❑ There are huge number of applications in the field of corporate sector, educational institutions, sports industry, e.t.c which involves the detection and tracking of an object to achieve the task
- ❑ For example some of the applications can be::
 - ✓ The detection and tracking of pedestrians to give information to drivers who drives vehicle on the road to avoid as many accidents as possible
 - ✓ The detection and tracking of suspicious person to achieve the security measures
 - ✓ The detection and tracking of old persons or babies in the houses to monitor their activities and includes an alarming system to inform the other house members in case of any dangerous situation
 - ✓ The detection and movement of instructor in front of the white board in case you are developing a system to extract the information from the white board into handouts from the video lectures e.t.c



Fingerprint Recognition [13, 14]

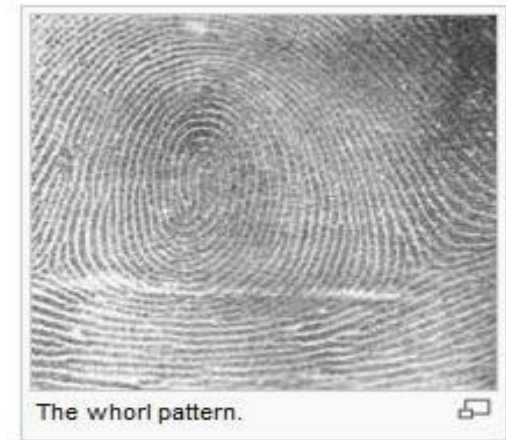
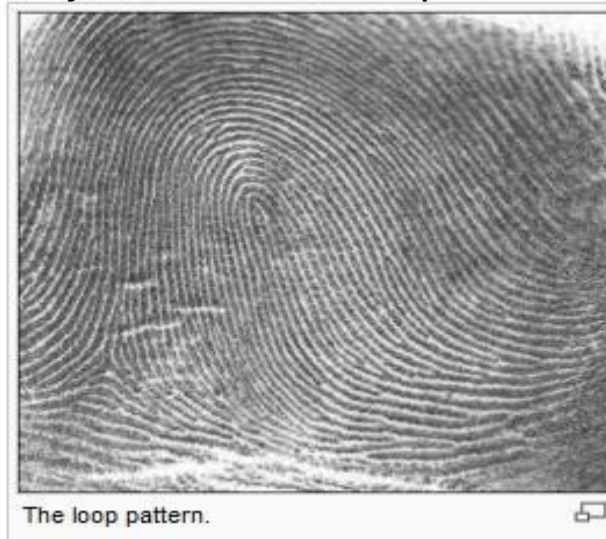
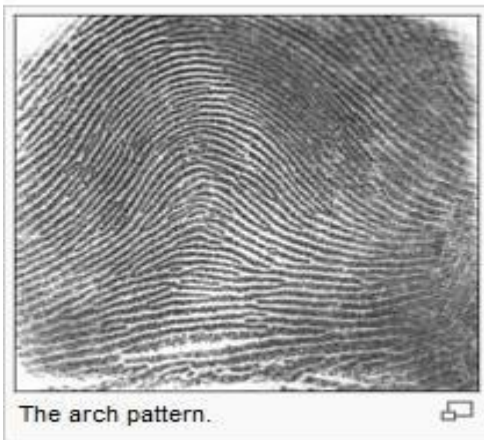


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Patterns

The three basic patterns of fingerprint ridges are the arch, loop, and whorl:

- ✓ arch: The ridges enter from one side of the finger, rise in the center forming an arc, and then exit the other side of the finger.
- ✓ loop: The ridges enter from one side of a finger, form a curve, and then exit on that same side.
- ✓ whorl: Ridges form circularly around a central point on the finger.



Face Detection

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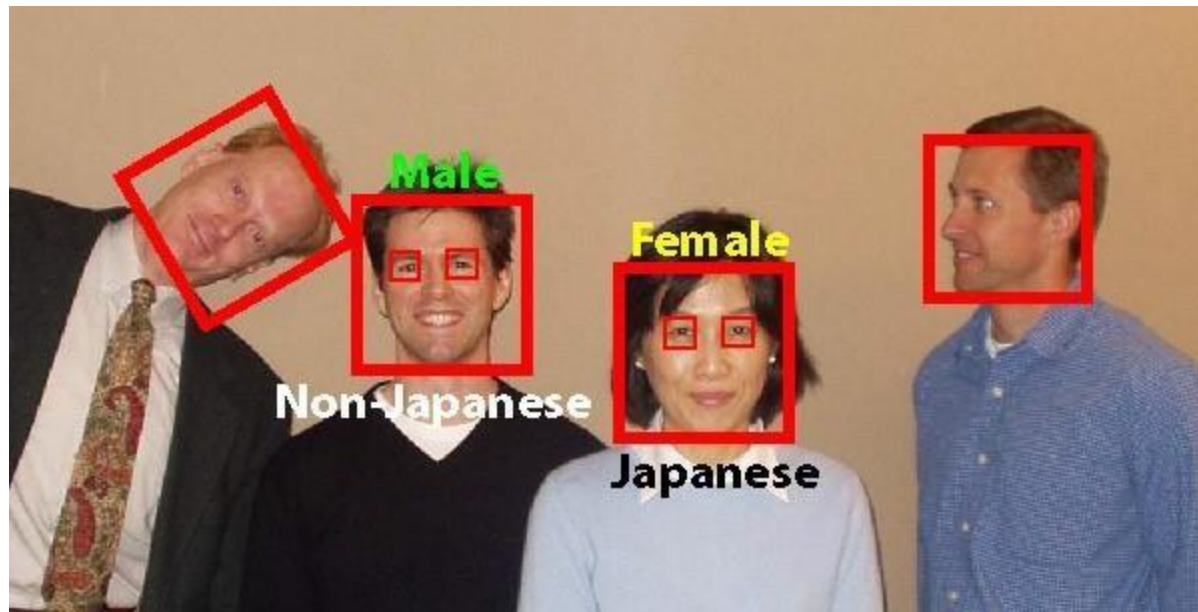
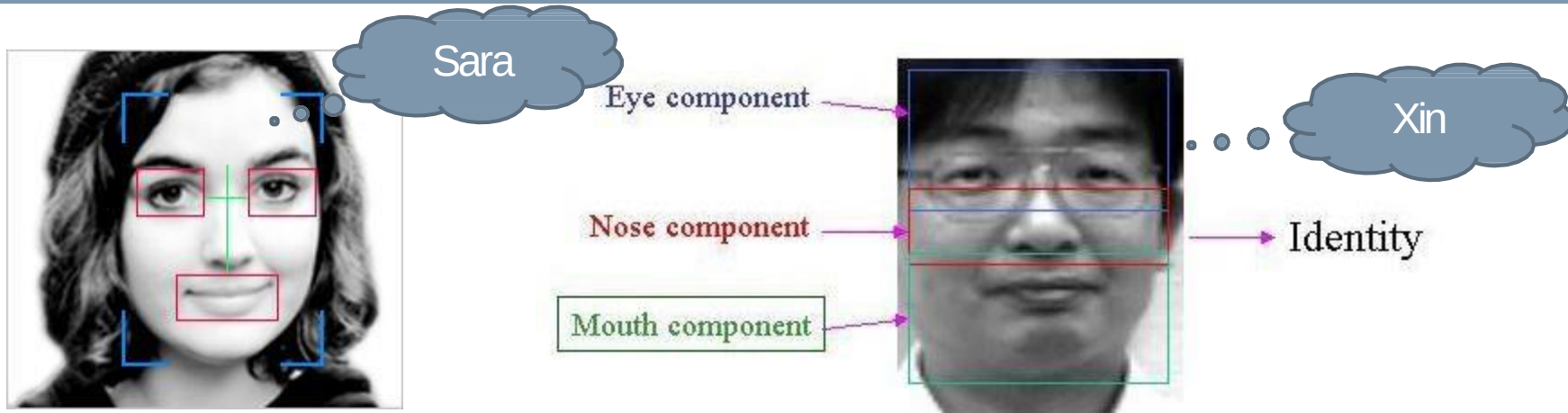
Face Tracking

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Face Recognition

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Personal Identification Using Iris Recognition

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HOW IRIS SCANNERS RECORD IDENTITIES

- ① Scanner reads from outer iris inwards to pupil edge
 - ② Scanner plots distinct markings on iris and maps unique shape
 - ③ After plotting many marks within the iris all data is saved to a database
 - ④ Other scanners will compare this data to verify individual identities
-
- The diagram shows a detailed illustration of a human eye. A grid of red and green lines is overlaid on the iris, representing the scanning process. The grid is centered on the pupil and extends to the outer edge of the iris. The grid lines are thicker at the outer edge and become thinner towards the pupil. The diagram is used to illustrate the four steps of the iris scanning process.

References

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2. <http://www.darvill.clara.net/emag/emaggamma.htm>
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14. <http://en.wikipedia.org/wiki/Minutiae>

For any query Feel Free to ask



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